

man crossed the 40-year mark, he had shown that very few ailments could harm him. We now proceed to a mathematical rephrasing of these arguments.

Philostratus in Monte Carlo: On the Difference Between Noise and Information

The wise man listens to meaning, the fool only gets the noise. The modern Greek poet C. P. Cavafy wrote a piece in 1915 after Philostratus' adage: *For the gods perceive things in the future, ordinary people things in the present, but the wise perceive things about to happen.* Cavafy wrote:

in their intense meditation the hidden sound of things approaching reaches them and they listen reverently while in the street outside the people hear nothing at all.

I thought hard and long on how to explain with as little mathematics as possible the difference between noise and meaning, and how to show why the time scale is important in judging an historical event. The Monte Carlo simulator can provide us with such an intuition. We will start with an example borrowed from the investment world (that is my profession), as it can be explained rather easily, but the concept can be used in any application.

Let us manufacture a happily retired dentist, living in a pleasant sunny town. We know *a priori* that he is an excellent investor, and that he will be expected to earn a return of 15% in excess of Treasury bills, with a 10% error rate per annum (what we call volatility). It means that out of 100 sample paths, we expect close to 68 of them to fall within a band of plus and minus 10% around the 15% excess return, i.e. between 5% and 25% (to be technical; the bell-shaped normal distribution has 68% of all observations falling between -1 and 1 standard deviations). It also means that 95 sample paths would fall between -5% and 35% .

Clearly, we are dealing with a very optimistic situation. The dentist builds for himself a nice trading desk in his attic, aiming to spend every business day there watching the market, while sipping decaffeinated cappuccino. He has an adventurous temperament, so he finds this activity more attractive than drilling the teeth of reluctant old little Park Avenue ladies.

He subscribes to a web-based service that supplies him with continuous prices, now to be obtained for a fraction of what he pays for his coffee. He puts his inventory of securities in his spreadsheet and can thus instantaneously monitor the value of his speculative portfolio. We are living in the era called that of connectivity.

A 15% return with a 10% volatility (or uncertainty) per annum translates into a 93% probability of making money in any given year. But seen at a narrow time scale, this translates into a mere 50.02% probability of making money over any given second as shown in Table 3.1. Over the very narrow time increment, the observation will reveal close to nothing. Yet the dentist's heart will not tell him that. Being emotional, he feels a pang with every loss, as it shows in red on his screen. He feels some pleasure when the performance is positive, but not in equivalent amount as the pain experienced when the performance is negative.

At the end of every day the dentist will be emotionally drained. A minute-by-minute examination of his performance means that each day (assuming eight hours per day) he will have 241 pleasurable minutes against 239 unpleasurable ones. These amount to 60,688 and 60,271, respectively, per year. Now realize that if the unpleasurable minute is

Table 3.1 Probability of making money at different scales.

Scale	Probability
1 year	93%
1 quarter	77%
1 month	67%
1 day	54%
1 hour	51.3%
1 minute	50.17%
1 second	50.02%